

Matica $[f]_B^B$ bude tvoriť:

$$\begin{pmatrix} \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}} & & \\ & \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}} & \\ & & \boxed{\begin{smallmatrix} 0 & 1 \\ 0 & 1 \end{smallmatrix}} \end{pmatrix}$$

Keďže je matica v Jordánovom tvare, vieme ju jednoducho umocniť, t.j. je umocňujeme jednotlivé Jordánove bloky.

$$J_{0,5}: J^2 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad J_{0,3}: J^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad J^{2,2} = 0_{2 \times 2}$$

$$J^3 = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad J^{2,3} = 0_{3 \times 3}$$

$$J^4 = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$J^{\geq 5} = 0_{5 \times 5}$$

Z toho plynie, že nám stačí zistiť $\text{Ker}(f^i)$ a $\text{Im}(f^i)$ pre $i \leq 4$, keďže pre väčšie i by $[f]_B^B$ bola nulová matica.

$i=1$:

$$\text{Ker}([f]_B^B) = \text{Ker} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = [\text{Span}\{e_1, e_6, e_9\}]_B = \text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9\}$$

$$\text{Im}([f]_B^B) = \text{Im} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = [\text{Span}\{e_1, e_2, e_3, e_4, e_6, e_7, e_9\}]_B = \text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_6, \vec{b}_7, \vec{b}_9\}$$

$i=2$:

$$\text{Ker}([f^2]_B^B) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = \text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_6, \vec{b}_7, \vec{b}_9, \vec{b}_{10}\}$$

$$\text{Im}([f^2]_B^B) = \text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6\}$$

$i=3$:

$$\text{Ker}([f^3]_B^B) = \text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6, \vec{b}_7, \vec{b}_9, \vec{b}_{10}\}$$

$$\text{Im}([f^3]_B^B) = \text{Span}\{\vec{b}_1, \vec{b}_2\}$$

$i=4$:

$$\text{Ker}([f^4]_B^B) = \text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_6, \vec{b}_7, \vec{b}_9, \vec{b}_{10}\}$$

$$\text{Im}([f^4]_B^B) = \text{Span}\{\vec{b}_1\}$$

$$i \geq 5: \text{Ker}([f^i]_B^B) = \mathbb{R}^{10}$$

$$\text{Im}([f^i]_B^B) = \{\vec{0}\}$$

Potom máme:	j	i	$\text{Ker } f^i \cap \text{Im } f^j$	$\dim(\text{Im}(f^j) \cap \text{Ker}(f^i))$
	1	1	$\text{Span}\{\vec{b}_1, \vec{b}_6, \vec{b}_9\}$	3
	1	2	$\text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_6, \vec{b}_7, \vec{b}_9\}$	5
	1	3	$\text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6, \vec{b}_7, \vec{b}_9\}$	6
	1	4	$\text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_4, \vec{b}_6, \vec{b}_7, \vec{b}_9\}$	7
	2	1	$\text{Span}\{\vec{b}_1, \vec{b}_6\}$	2
	2	2	$\text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_6\}$	3
	2	3, 4	$\text{Span}\{\vec{b}_1, \vec{b}_2, \vec{b}_3, \vec{b}_6\}$	4
	3	1	$\text{Span}\{\vec{b}_1\}$	1
	3	2, 3, 4	$\text{Span}\{\vec{b}_1, \vec{b}_2\}$	2
	4	1, 2, 3, 4	$\text{Span}\{\vec{b}_1\}$	1

Ak $j \geq 5$, potom $\text{Im } f^j = \{\vec{0}\}$ a $\text{Im } f^j \cap \text{Ker } f^i = \{\vec{0}\}$

Ak $i \geq 5$, potom $\text{Ker } f^i = \mathbb{R}^{10}$ a $\text{Im } f^j \cap \text{Ker } f^i = \text{Im } f^j$.